DS3 Data Science for Developing Scholars in Down Syndrome Research

Introduction to Linear Regression Modeling and Hypothesis Testing

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Linda Crnic Institute for Down Syndrome

Data Science for Developing Scholars in Down Syndrome Research (DS3) 2026

INCLUDE INvestigation of Co-occurring conditions across the Lifespan to Understand Down syndromE

Linda Crnic Institute for Down Syndrome
UNIVERSITY OF COLORADO ANSCHUTZ MEDICAL CAMPUS

GLOBAL DOWN SYNDROME FOUNDATION®

Links for this session

Course materials

DS3 2026 repository on GitHub

DS3 Linear Regression Exercise

DS3 HTP Linear Regression Example

The Data Science Workflow

Import data

↓

Data Cleaning / Tidying

↓

Exploratory Data Analysis
(incl. visualization)

↓

Statistical Analysis

↓

Visualization of results

↓

Interpretation

↓

Biological Insight

Today's exercises focus on statistical analysis, results visualization, and interpretation

Introduction to linear regression modeling

- Linear regression can be used to model a linear relationships between a **response variable** and one (simple regression) or more (multiple regression) **predictor variables**
- The linear relationship (ie straight line) can be described in the form:

$$y = mx + c$$

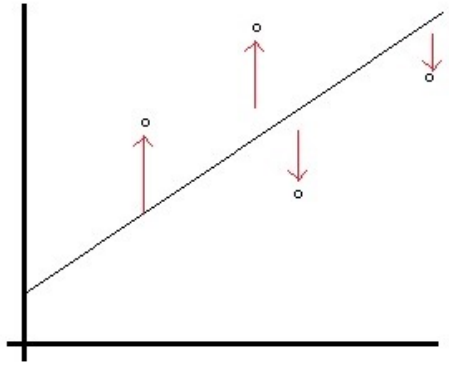
where y is the response (dependent) variable

m is the gradient (slope) (aka beta 2)

x is the predictor (independent) variable

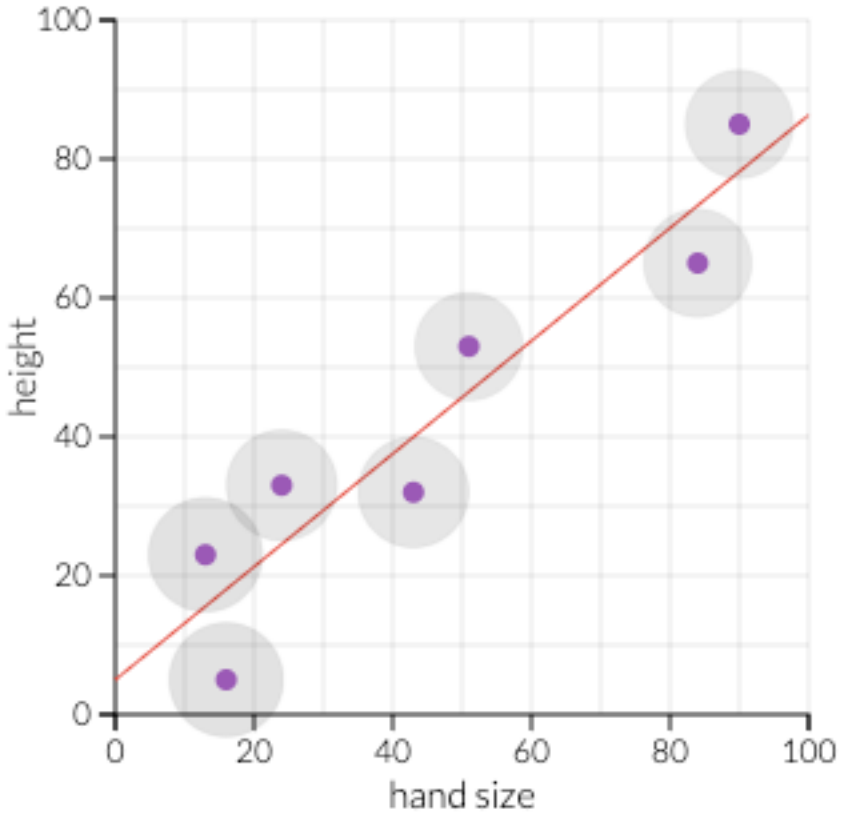
c is the intercept (aka beta 1)

- The Ordinary Least Squares (OLS) approach finds the **line of best fit** through the data points by minimizing the variance (the sum of squares of the errors)



https://www.r-bloggers.com/2017/07/ordinary-least-squares-ols-linear-regression-in-r/
https://scientificallysound.org/2017/06/08/t-test-as-linear-models-r/
https://lindeloev.github.io/tests-as-linear/

Linear regression modeling: Ordinary Least Squares



Beta 1 - The y-intercept of the regression line.

5.00 + 0.81 * hand size = height

Beta 2 - The slope of the regression line.

Source: https://setosa.io/ev/ordinary-least-squares-regression/index.html

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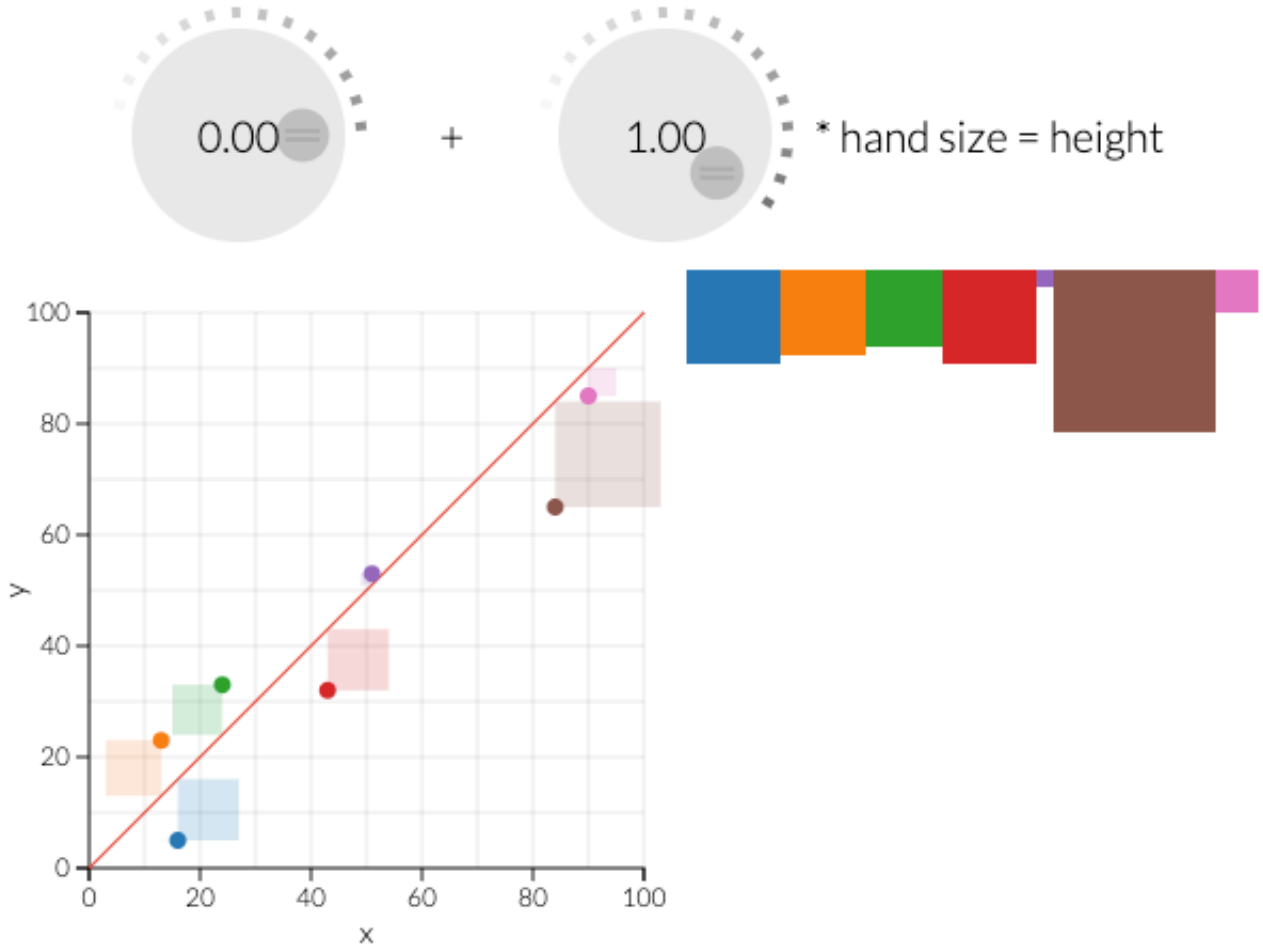

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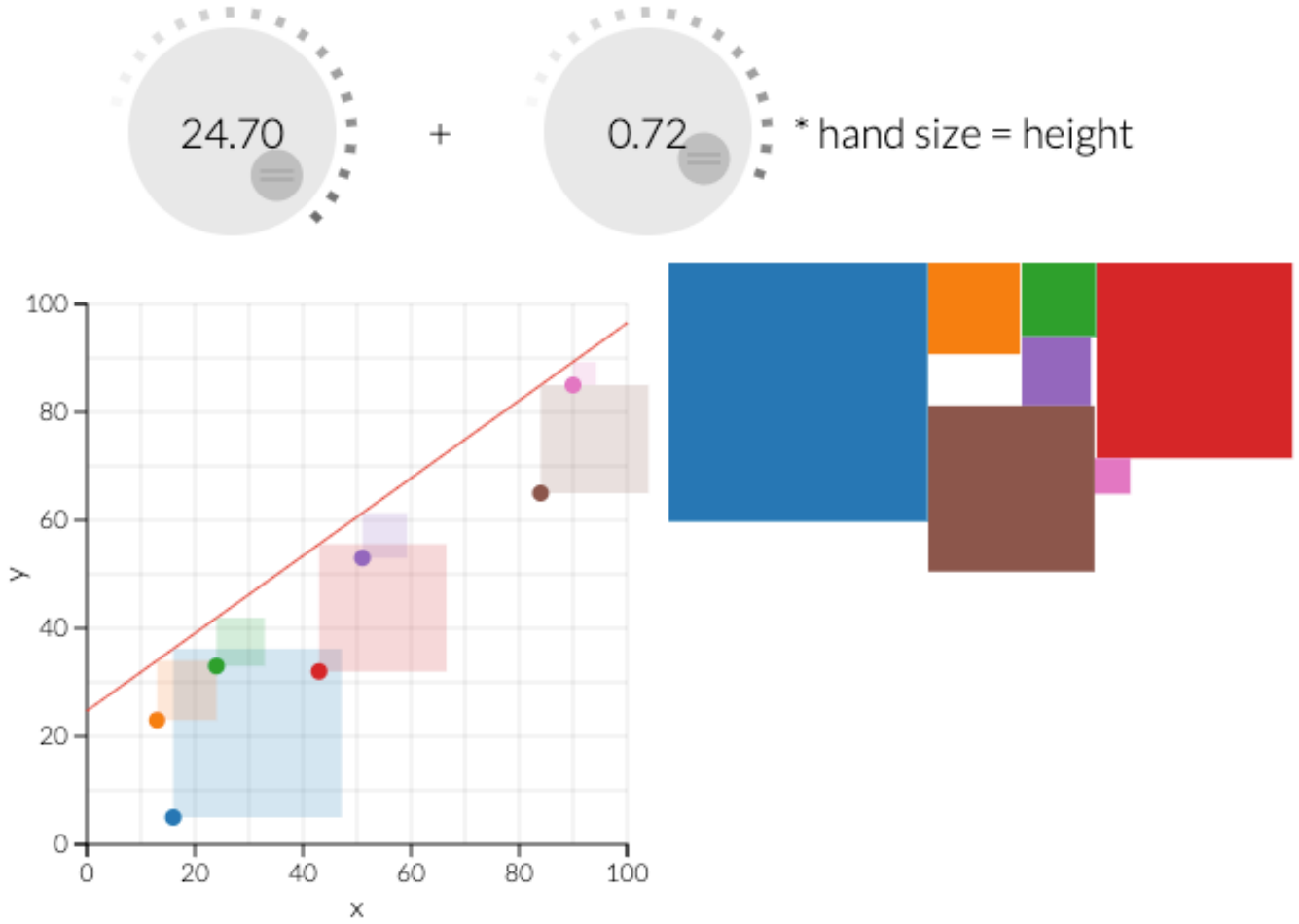

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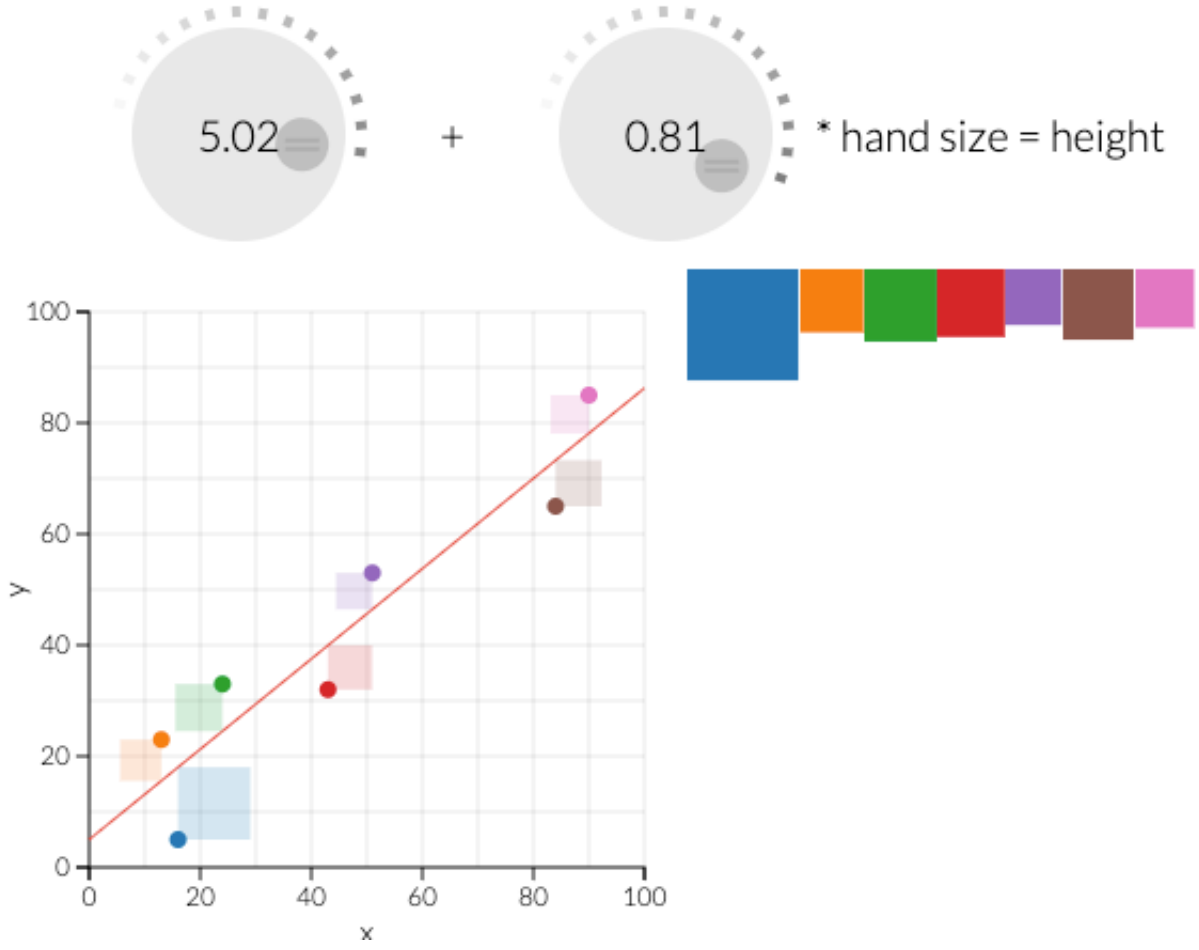

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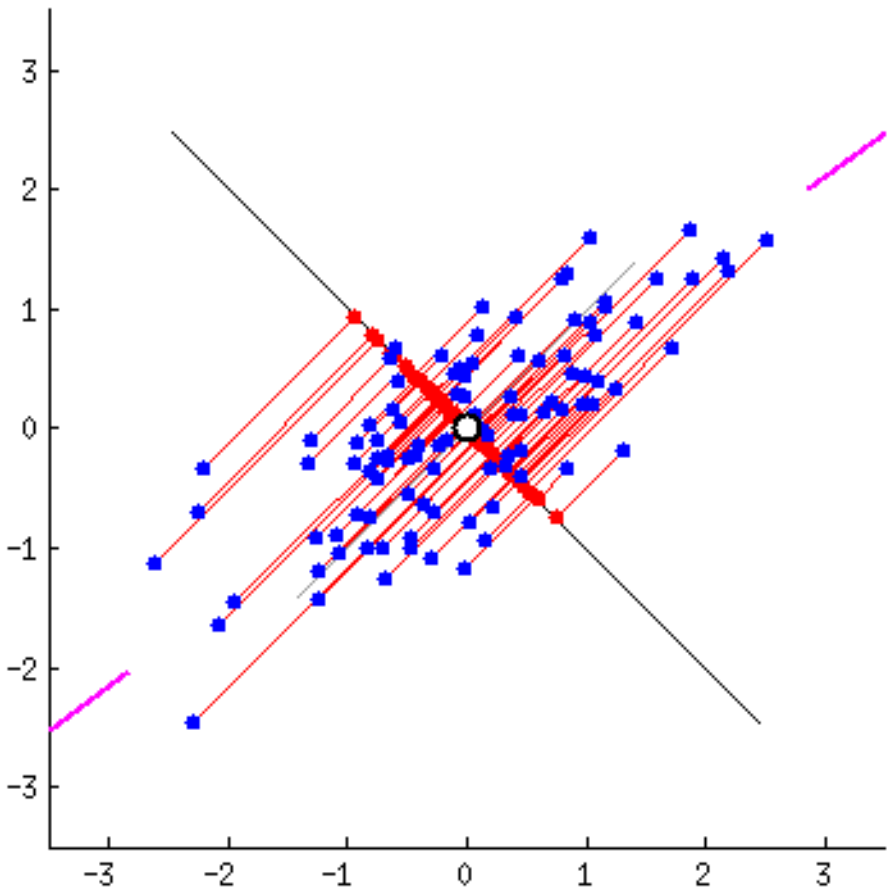
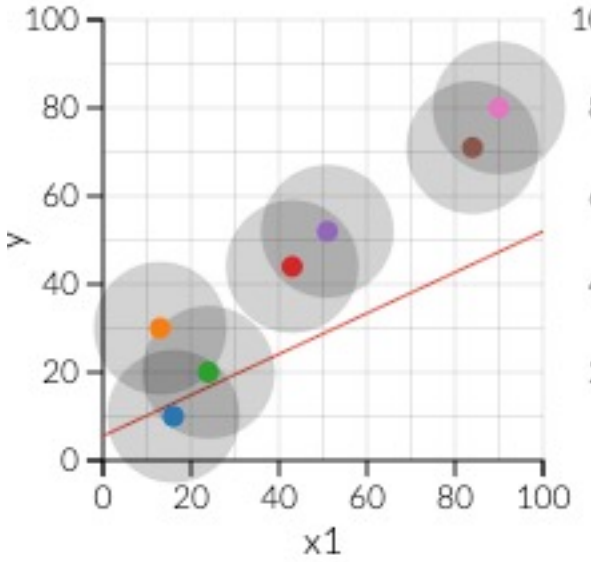
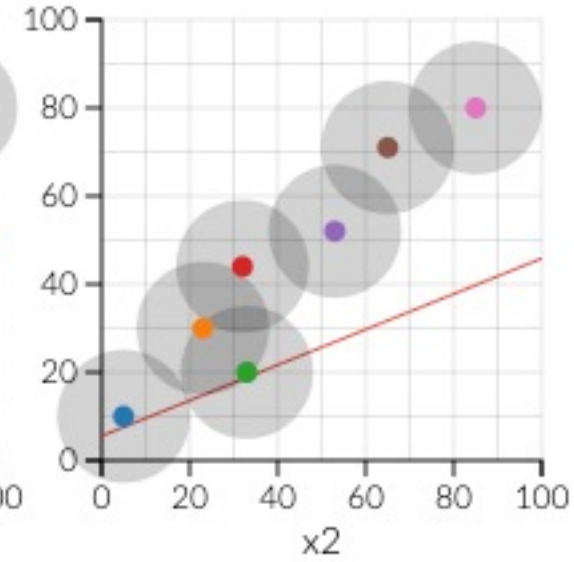


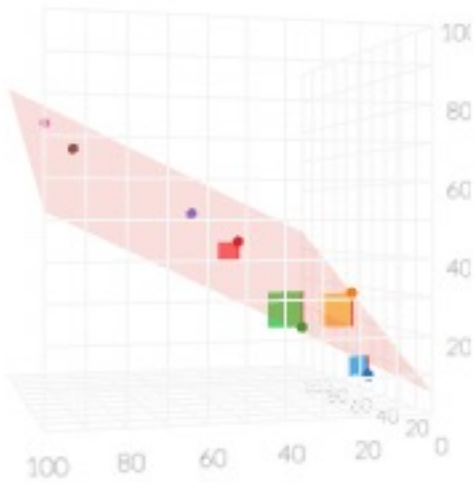
Source: stats.stackexchange.com

Linear regression modeling: Ordinary Least Squares

- Often, we have more than one independent variable
- Errors are now relative to a plane in 3D space (or greater)







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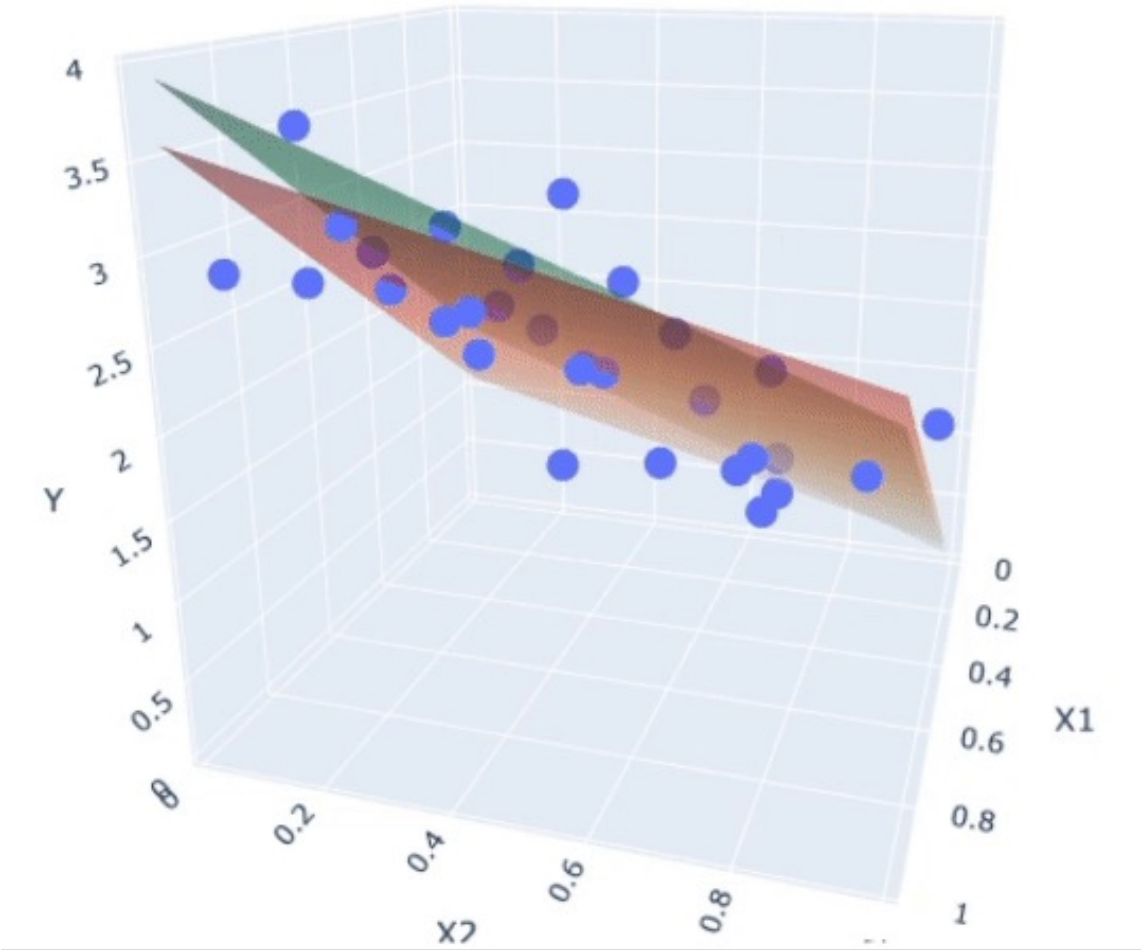


Source: https://medium.com/@MerelyAI/visualizing-ordinary-least-squares-regression-in-3d-0ce4b7372fb8

Linear regression modeling: Assumptions

Assumptions of linear regression

1. Linearity: The relationship between the independent variable and the study variable is assumed to be linear.
2. Homoscedasticity: The error term (ε) is assumed to have a constant variance.
3. Independence: We assume observations are independent of each other.
4. Normality: Observations are assumed to have a normal distribution.

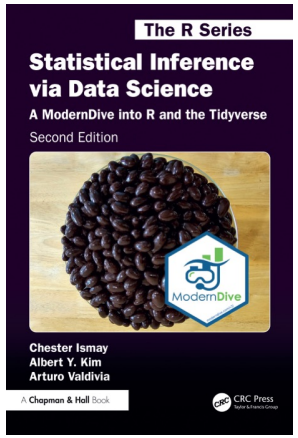
Significant deviation from these assumptionss may invalidate your results!

Linear regression can be used for both <u>prediction</u> and <u>hypothesis testing</u>

Introduction to Hypothesis Testing and Statistical Inference

Often, we seek to make claims about population parameters with some measure of the plausibility (eg p-value)

- A **hypothesis test** consists of a test between two competing hypotheses:
  1. H_0, the **null** hypothesis eg "There is **NO difference** in the means of Group A and Group B"
  2. H_A, the **alternative** hypothesis eg "The means of Group A and Group B **ARE different**"
- A **test statistic** is a *point estimate/sample statistic* formula used for hypothesis testing (eg mean)
- The **null distribution** is the sampling distribution of the test statistic *assuming the null hypothesis* H0H0 *is true*
- A p-**value** is the probability of obtaining a test statistic just as extreme as or more extreme than the observed test statistic *assuming the null hypothesis* H_0 *is true*.
- If the p-value is less than the **significance level** (α), we reject the null hypothesis H_0, otherwise we fail to reject H_0



https://moderndive.com/v2/index.html

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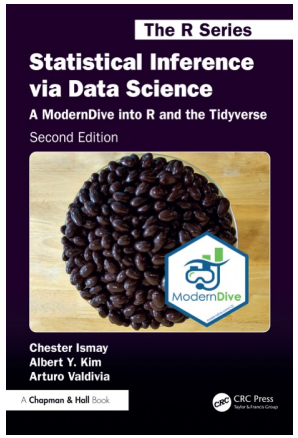


In the case of a linear regression model, we can:

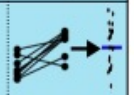
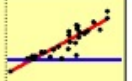
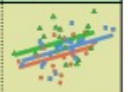
- Determine whether a predictor variable has a statistically significant relationship with an outcome variable.
- Estimate the difference between two or more groups.

https://moderndive.com/v2/index.html

Common statistical tests are linear models

Last updated: 28 June, 2019.Also check out the Python version!

See worked examples and more details at the accompanying notebook: https://lindeloev.github.io/tests-as-linear

	Common name	Built-in function in R	Equivalent linear model in R	Exact?	The linear model in words	Icon
Simple regression: lm(y ~ 1 + x)	**y is independent of x** P: One-sample t-test N: Wilcoxon signed-rank	t.test(y) wilcox.test(y)	lm(y ~ 1) lm(signed_rank(y) ~ 1)	✓ for N >14	One number (intercept, i.e., the mean) predicts **y**. - (Same, but it predicts the *signed rank* of **y**.)	
	P: Paired-sample t-test N: Wilcoxon matched pairs	t.test(y_1, y_2, paired=TRUE) wilcox.test(y_1, y_2, paired=TRUE)	lm(y_2 - y_1 ~ 1) lm(signed_rank(y_2 - y_1) ~ 1)	✓ for N >14	One intercept predicts the pairwise $\mathbf{y_2}$-$\mathbf{y_1}$ differences. - (Same, but it predicts the *signed rank* of $\mathbf{y_2}$-$\mathbf{y_1}$.)	
	y ~ continuous x P: Pearson correlation N: Spearman correlation	cor.test(x, y, method='Pearson') cor.test(x, y, method='Spearman')	lm(y ~ 1 + x) lm(rank(y) ~ 1 + rank(x))	✓ for N >10	One intercept plus **x** multiplied by a number (slope) predicts **y**. - (Same, but with *ranked* **x** and **y**)	
	y ~ discrete x P: Two-sample t-test P: Welch's t-test N: Mann-Whitney U	t.test(y_1, y_2, var.equal=TRUE) t.test(y_1, y_2, var.equal=FALSE) wilcox.test(y_1, y_2)	lm(y ~ 1 + G_2)[A] gls(y ~ 1 + G_2, weights=…[B])[A] lm(signed_rank(y) ~ 1 + G_2)[A]	✓ ✓ for N >11	An intercept for **group 1** (plus a difference if **group 2**) predicts **y**. - (Same, but with one variance *per group* instead of one common.) - (Same, but it predicts the *signed rank* of **y**.)	
Multiple regression: lm(y ~ 1 + x_1 + x_2 +…)	P: One-way ANOVA N: Kruskal-Wallis	aov(y ~ group) kruskal.test(y ~ group)	lm(y ~ 1 + G_2 + G_3 +…+ G_N)[A] lm(rank(y) ~ 1 + G_2 + G_3 +…+ G_N)[A]	✓ for N >11	An intercept for **group 1** (plus a difference if group ≠ 1) predicts **y**. - (Same, but it predicts the *rank* of **y**.)	
	P: One-way ANCOVA	aov(y ~ group + x)	lm(y ~ 1 + G_2 + G_3 +…+ G_N + x)[A]	✓	- (Same, but plus a slope on **x**.) *Note: this is discrete AND continuous. ANCOVAs are ANOVAs with a continuous x.*	
	P: Two-way ANOVA	aov(y ~ group * sex)	lm(y ~ 1 + G_2 + G_3 +…+ G_N + S_2 + S_3 +…+ S_K + G_2*S_2 + G_3*S_3 + … + G_N*S_K)	✓	Interaction term: changing **sex** changes the **y ~ group** parameters. *Note: $\mathbf{G_{2\,to\,N}}$ is an indicator (0 or 1) for each non-intercept levels of the **group** variable. Similarly for $\mathbf{S_{2\,to\,K}}$ for sex. The first line (with G_i) is main effect of group, the second (with S_i) for sex and the third is the **group × sex** interaction. For two levels (e.g. male/female), line 2 would just be "S_2" and line 3 would be S_2 multiplied with each G_i.*	*[Coming]*
	Counts ~ discrete x N: Chi-square test	chisq.test(groupXsex_table)	**Equivalent log-linear model** glm(y ~ 1 + G_2 + G_3 + … + G_N + S_2 + S_3+ … + S_K + G_2*S_2 + G_3*S_3 +…+ G_N*S_K, family=…)[A]	✓	Interaction term: (Same as Two-way ANOVA.) *Note: Run glm using the following arguments: `glm(model, family=poisson())` As linear-model, the Chi-square test is $\log(y_i) = \log(N) + \log(\alpha_i) + \log(\beta_j) + \log(\alpha_i\beta_j)$ where α_i and β_j are proportions. See more info in the accompanying notebook.*	*Same as Two-way ANOVA*
	N: Goodness of fit	chisq.test(y)	glm(y ~ 1 + G_2 + G_3 +…+ G_N, family=…)[A]	✓	(Same as One-way ANOVA and see Chi-Square note.)	*1W-ANOVA*

List of common parametric (P) non-parametric (N) tests and equivalent linear models. The notation y ~ 1 + x is R shorthand for y = 1·b + a·x which most of us learned in school. Models in similar colors are highly similar, but really, notice how similar they *all* are across colors! For non-parametric models, the linear models are reasonable approximations for non-small sample sizes (see "Exact" column and click links to see simulations). Other less accurate approximations exist, e.g., Wilcoxon for the sign test and Goodness-of-fit for the binomial test. The signed rank function is `signed_rank = function(x) sign(x) * rank(abs(x))`. The variables G_i and S_i are "dummy coded" indicator variables (either 0 or 1) exploiting the fact that when Δx = 1 between categories the difference equals the slope. Subscripts (e.g., G_2 or y_1) indicate different columns in data. lm requires long-format data for all non-continuous models. All of this is exposed in greater detail and worked examples at https://lindeloev.github.io/tests-as-linear.

[A] See the note to the two-way ANOVA for explanation of the notation.
[B] Same model, but with one variance per group: `gls(value ~ 1 + G2, weights = varIdent(form = ~1|group), method="ML")`.

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Jonas Kristoffer Lindeløv
https://lindeloev.net

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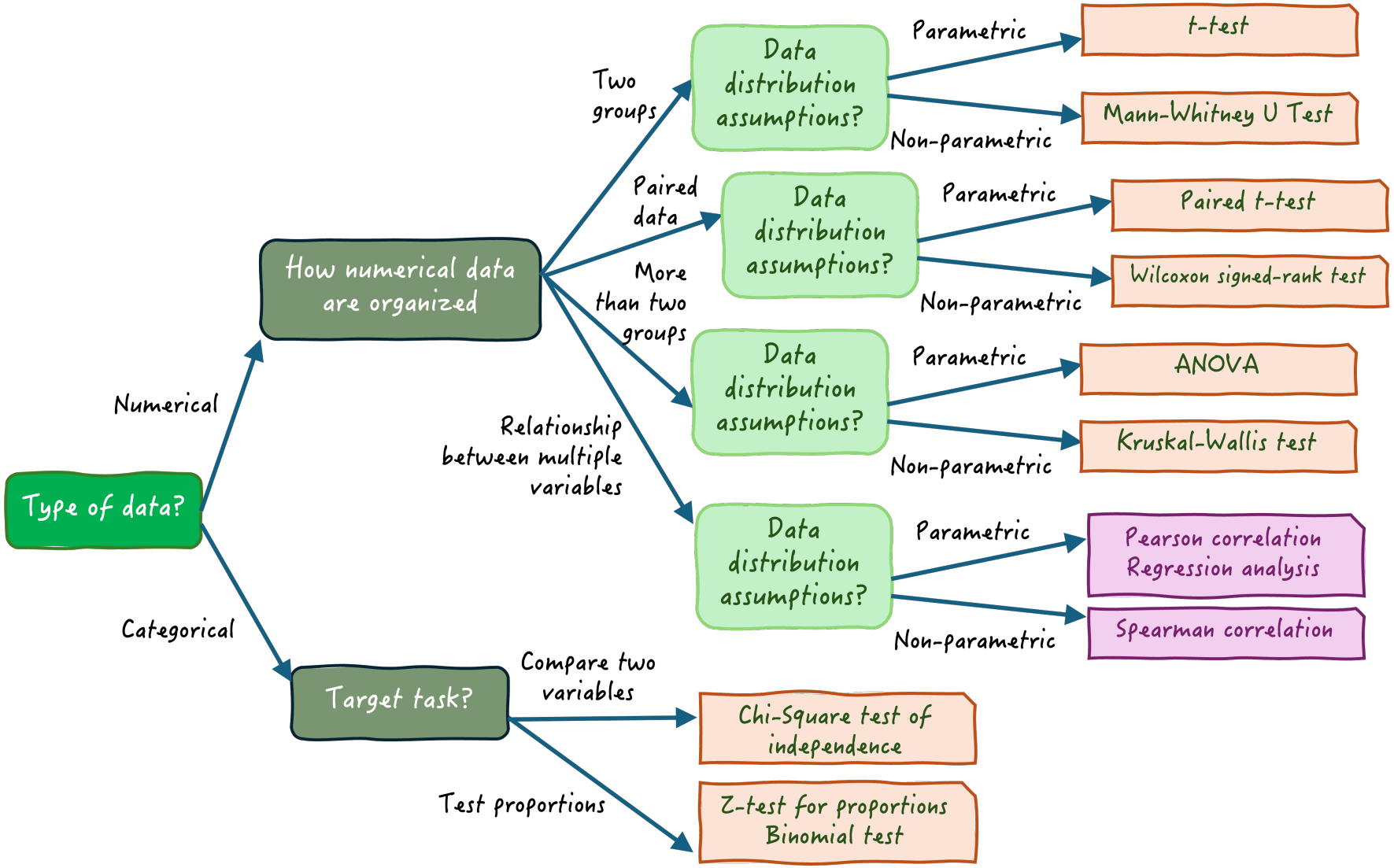
Choosing a Statistical Test (not exhaustive)

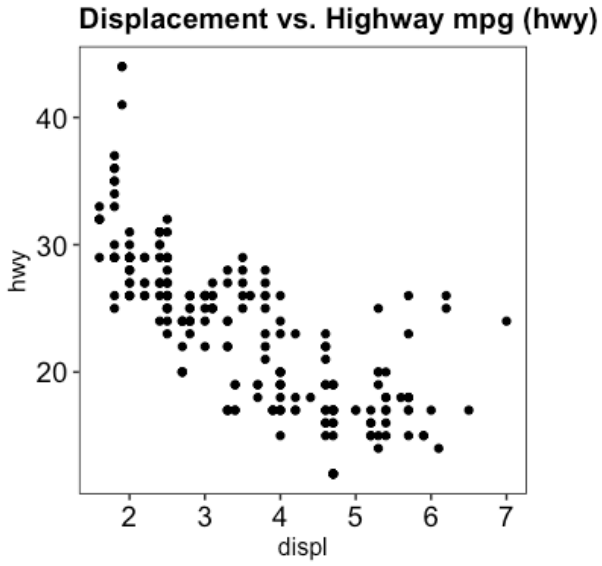


https://www.statology.org/choosing-the-right-statistical-test-a-decision-tree-approach/

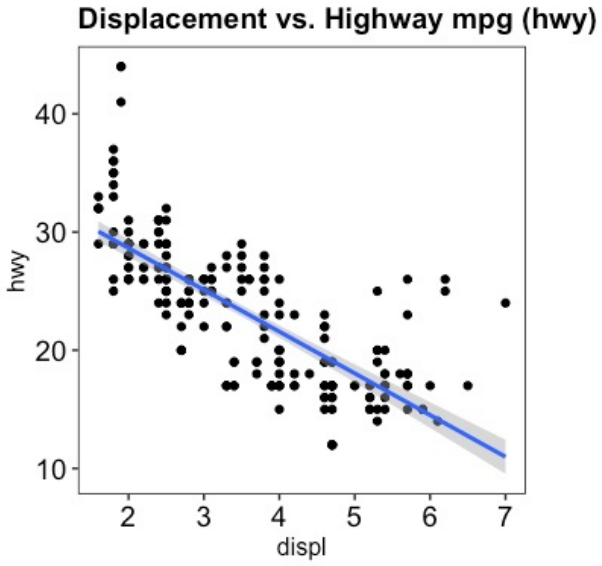
Linear regression modeling in R

Explore the data
How do the response and predictor relate to each other?
Does the relationship appear linear?
Consider transforming the data

```
mpg %>%
    ggplot(aes(displ, hwy)) +
    geom_point() +
    theme(aspect.ratio = 1) +
    labs(title = "Displacement vs. Highway mpg (hwy)")
```




```
mpg %>%
    ggplot(aes(displ, hwy)) +
    geom_point() +
    geom_smooth(method = "lm") +
    theme(aspect.ratio = 1) +
    labs(title = "Displacement vs. Highway mpg (hwy)")
```



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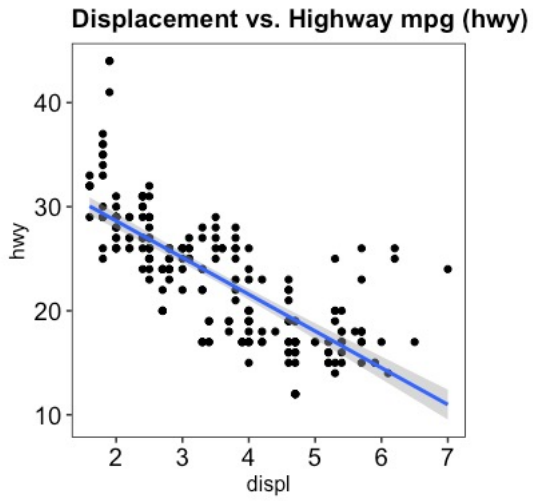
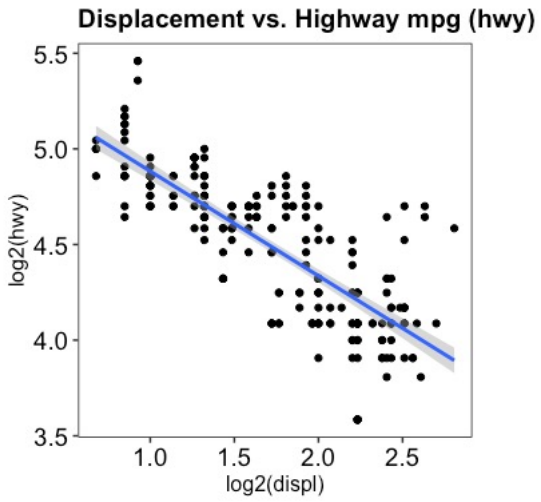



```
mpg %>%
    ggplot(aes(log2(displ), log2(hwy))) +
    geom_point() +
    geom_smooth(method = "lm") +
    theme(aspect.ratio = 1) +
    labs(title = "Displacement vs. Highway mpg (hwy)")
```




```
> lm(hwy ~ displ, data = mpg) %>% summary()

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  35.6977     0.7204   49.55   <2e-16 ***
displ        -3.5306     0.1945  -18.15   <2e-16 ***
---
Residual standard error: 3.836 on 232 degrees of freedom
Multiple R-squared:  0.5868,	Adjusted R-squared:  0.585
F-statistic: 329.5 on 1 and 232 DF,  p-value: < 2.2e-16
```

```
> lm(log2(hwy) ~ log2(displ), data = mpg) %>% summary()

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.43041    0.04853  111.90   <2e-16 ***
log2(displ) -0.54716    0.02726  -20.07   <2e-16 ***
---
Residual standard error: 0.2276 on 232 degrees of freedom
Multiple R-squared:  0.6345,	Adjusted R-squared:  0.6329
F-statistic: 402.8 on 1 and 232 DF,  p-value: < 2.2e-16
```

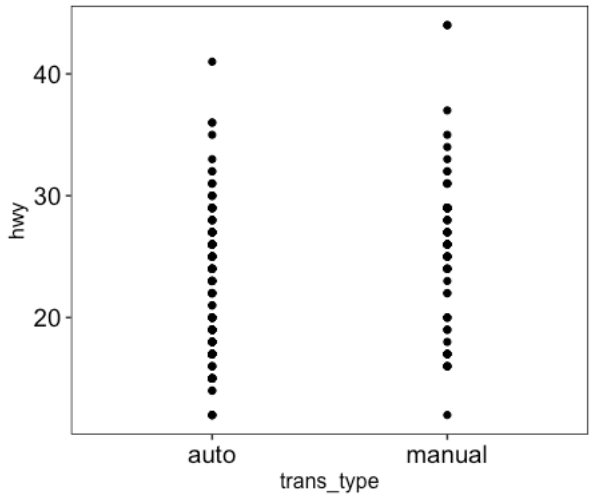
We are usually NOT interested in the coefficient (estimate) for the intercept

We are testing against the *null hypothesis that the coefficient (estimate) for a variable of interest is zero*

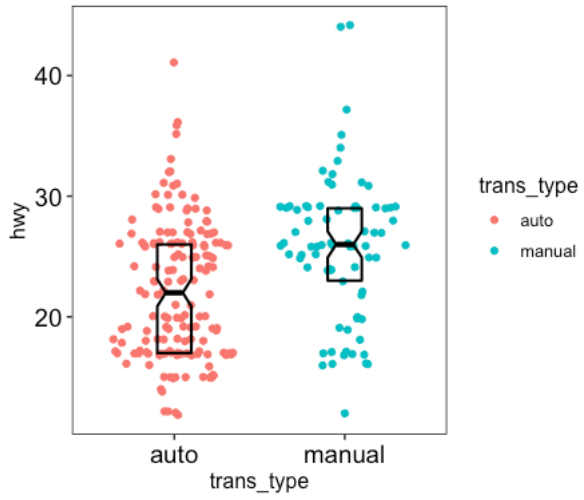
Linear regression modeling in R

```
> mpg %>% mutate(trans_type = str_extract(trans, "\\w+(?=\\()"))
# A tibble: 234 × 12
   manufacturer model      displ  year   cyl trans      drv     cty   hwy fl    class   trans_type
   <chr>        <chr>      <dbl> <int> <int> <chr>      <chr> <int> <int> <chr> <chr>   <chr>
 1 audi         a4           1.8  1999     4 auto(l5)   f        18    29 p     compact auto
 2 audi         a4           1.8  1999     4 manual(m5) f        21    29 p     compact manual
 3 audi         a4           2    2008     4 manual(m6) f        20    31 p     compact manual
 4 audi         a4           2    2008     4 auto(av)   f        21    30 p     compact auto
 5 audi         a4           2.8  1999     6 auto(l5)   f        16    26 p     compact auto
 6 audi         a4           2.8  1999     6 manual(m5) f        18    26 p     compact manual
 7 audi         a4           3.1  2008     6 auto(av)   f        18    27 p     compact auto
 8 audi         a4 quattro   1.8  1999     4 manual(m5) 4        18    26 p     compact manual
 9 audi         a4 quattro   1.8  1999     4 auto(l5)   4        16    25 p     compact auto
10 audi         a4 quattro   2    2008     4 manual(m6) 4        20    28 p     compact manual
# … with 224 more rows
```

```
mpg %>%
    mutate(trans_type = str_extract(trans, "\\w+(?=\\()")) %>%
    ggplot(aes(trans_type, hwy)) +
    geom_point()
```




```
mpg %>%
    mutate(trans_type = str_extract(trans, "\\w+(?=\\()")) %>%
    ggplot(aes(trans_type, hwy, color = trans_type)) +
    geom_sina()
```



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```
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    ggplot(aes(trans_type, hwy, color = trans_type)) +
    geom_sina() +
    geom_boxplot(notch = TRUE, varwidth = FALSE, outlier.shape = NA, coef =
FALSE, width = 0.2, color = "black", fill = "transparent", size = 0.75) +
    geom_smooth(method = "lm", aes(group = "1"))
```
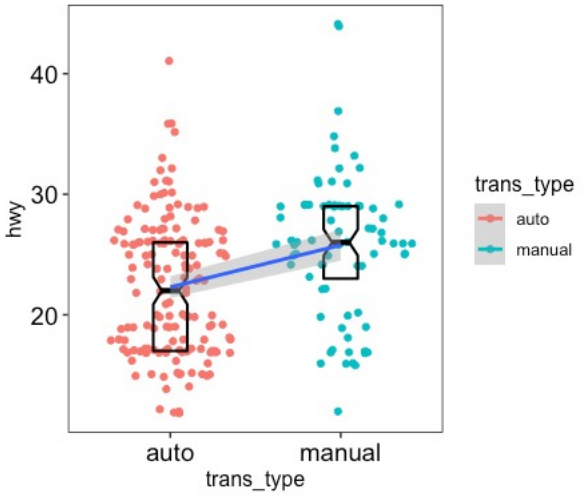
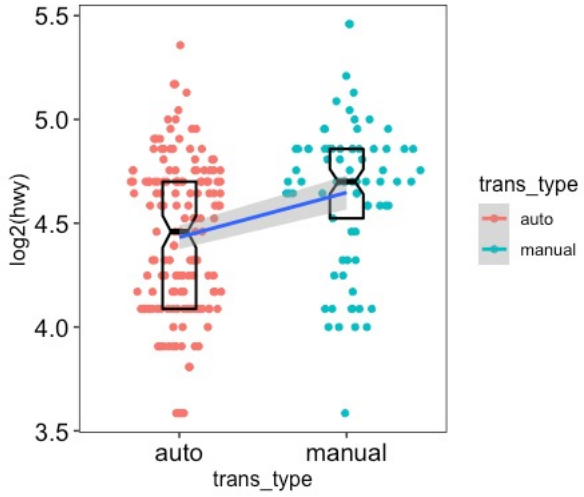




```
mpg %>%
    mutate(trans_type = str_extract(trans, "\\w+(?=\\()")) %>%
    ggplot(aes(trans_type, log2(hwy), color = trans_type)) +
    geom_sina() +
    geom_boxplot(notch = TRUE, varwidth = FALSE, outlier.shape = NA, coef =
FALSE, width = 0.2, color = "black", fill = "transparent", size = 0.75) +
    geom_smooth(method = "lm", aes(group = "1"))
```

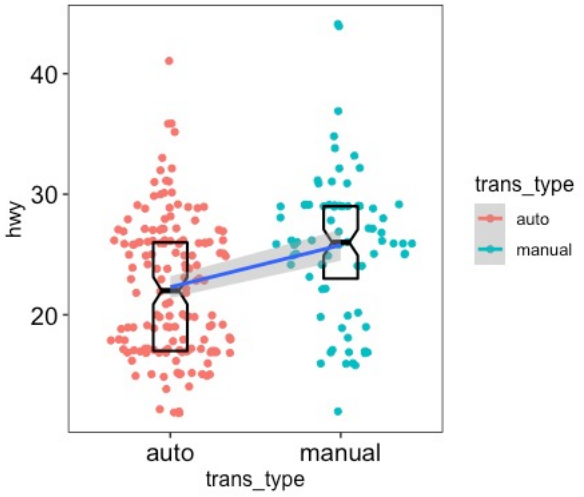



```
> mpg %>%
+     mutate(trans_type = str_extract(trans, "\\w+(?=\\()")) %>%
+     lm(hwy ~ trans_type, data = .) %>%
+     summary()

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       22.2930     0.4578  48.696  < 2e-16 ***
trans_typemanual   3.4862     0.7981   4.368 1.89e-05 ***
---
Residual standard error: 5.736 on 232 degrees of freedom
Multiple R-squared:  0.076,	Adjusted R-squared:  0.07202
F-statistic: 19.08 on 1 and 232 DF,  p-value: 1.888e-05
```

```
> mpg %>%
+     mutate(trans_type = str_extract(trans, "\\w+(?=\\()")) %>%
+     lm(log2(hwy) ~ trans_type, data = .) %>%
+     summary()

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       4.43244    0.02893 153.217  < 2e-16 ***
trans_typemanual  0.21554    0.05043   4.274 2.81e-05 ***
---
Residual standard error: 0.3625 on 232 degrees of freedom
Multiple R-squared:  0.07299,	Adjusted R-squared:  0.06899
F-statistic: 18.27 on 1 and 232 DF,  p-value: 2.806e-05
```

Linear regression modeling in R

```
mpg %>%
    mutate(trans_type = str_extract(trans, "\\w+(?=\\()")) %>%
    ggplot(aes(trans_type, hwy, color = trans_type)) +
    geom_sina() +
    geom_boxplot(notch = TRUE, varwidth = FALSE, outlier.shape = NA, coef =
FALSE, width = 0.2, color = "black", fill = "transparent", size = 0.75) +
    geom_smooth(method = "lm", aes(group = "1"))
```




```
> mpg %>%
+     mutate(trans_type = str_extract(trans, "\\w+(?=\\()")) %>%
+     lm(hwy ~ trans_type, data = .) %>%
+     summary()

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       22.2930     0.4578  48.696  < 2e-16 ***
trans_typemanual   3.4862     0.7981   4.368 1.89e-05 ***
---
Residual standard error: 5.736 on 232 degrees of freedom
Multiple R-squared:  0.076,	Adjusted R-squared:  0.07202
F-statistic: 19.08 on 1 and 232 DF,  p-value: 1.888e-05
```

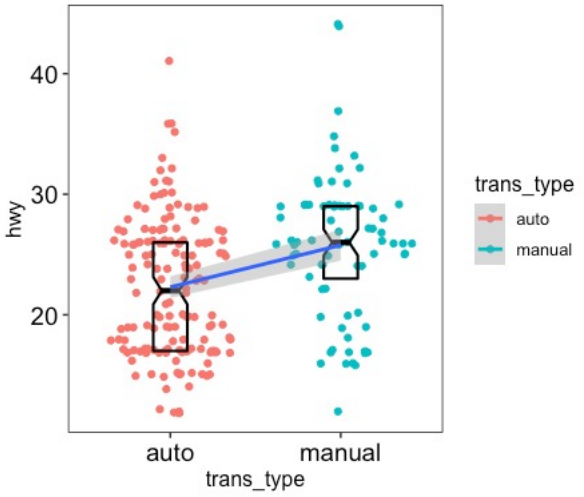
```
> mpg %>%
+     mutate(trans_type = str_extract(trans, "\\w+(?=\\()")) %>%
+     lm(log2(hwy) ~ trans_type, data = .) %>%
+     broom::tidy()
# A tibble: 2 × 5
  term             estimate std.error statistic   p.value
  <chr>               <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)         4.43     0.0289    153.   4.28e-235
2 trans_typemanual    0.216    0.0504      4.27 2.81e-  5
```

Simple linear regression gives (very) similar result to a t-test:

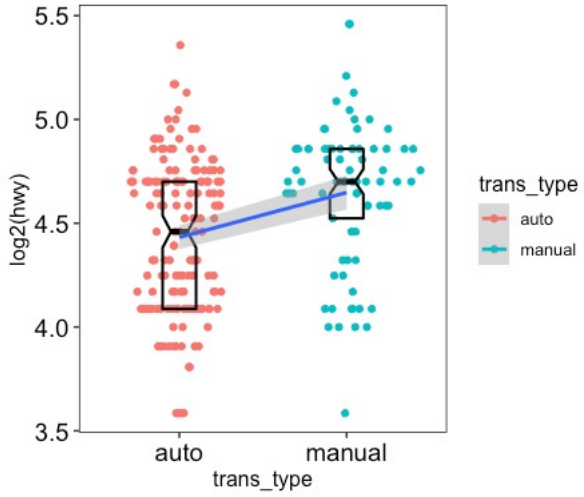
```
> mpg %>%
+     mutate(trans_type = str_extract(trans, "\\w+(?=\\()")) %>%
+     t.test(log2(hwy) ~ trans_type, data = .)

        Welch Two Sample t-test

data:  log2(hwy) by trans_type
t = -4.3483, df = 158.14, p-value = 2.448e-05
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.3134373 -0.1176359
sample estimates:
  mean in group auto mean in group manual
            4.432444             4.647981
```

Linear regression modeling in R

```
mpg %>%
    mutate(trans_type = str_extract(trans, "\\w+(?=\\()")) %>%
    ggplot(aes(trans_type, hwy, color = trans_type)) +
    geom_sina() +
    geom_boxplot(notch = TRUE, varwidth = FALSE, outlier.shape = NA, coef =
FALSE, width = 0.2, color = "black", fill = "transparent", size = 0.75) +
    geom_smooth(method = "lm", aes(group = "1"))
```




```
mpg %>%
    mutate(trans_type = str_extract(trans, "\\w+(?=\\()")) %>%
    ggplot(aes(trans_type, log2(hwy), color = trans_type)) +
    geom_sina() +
    geom_boxplot(notch = TRUE, varwidth = FALSE, outlier.shape = NA, coef =
FALSE, width = 0.2, color = "black", fill = "transparent", size = 0.75) +
    geom_smooth(method = "lm", aes(group = "1"))
```




```
> mpg %>%
+     mutate(trans_type = str_extract(trans, "\\w+(?=\\()")) %>%
+     lm(hwy ~ trans_type, data = .) %>%
+     summary()

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       22.2930     0.4578  48.696  < 2e-16 ***
trans_typemanual   3.4862     0.7981   4.368 1.89e-05 ***
---
Residual standard error: 5.736 on 232 degrees of freedom
Multiple R-squared:  0.076,	Adjusted R-squared:  0.07202
F-statistic: 19.08 on 1 and 232 DF,  p-value: 1.888e-05
```

```
> mpg %>%
+     mutate(trans_type = str_extract(trans, "\\w+(?=\\()")) %>%
+     lm(log2(hwy) ~ trans_type, data = .) %>%
+     summary()

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       4.43244    0.02893 153.217  < 2e-16 ***
trans_typemanual  0.21554    0.05043   4.274 2.81e-05 ***
---
Residual standard error: 0.3625 on 232 degrees of freedom
Multiple R-squared:  0.07299,	Adjusted R-squared:  0.06899
F-statistic: 18.27 on 1 and 232 DF,  p-value: 2.806e-05
```

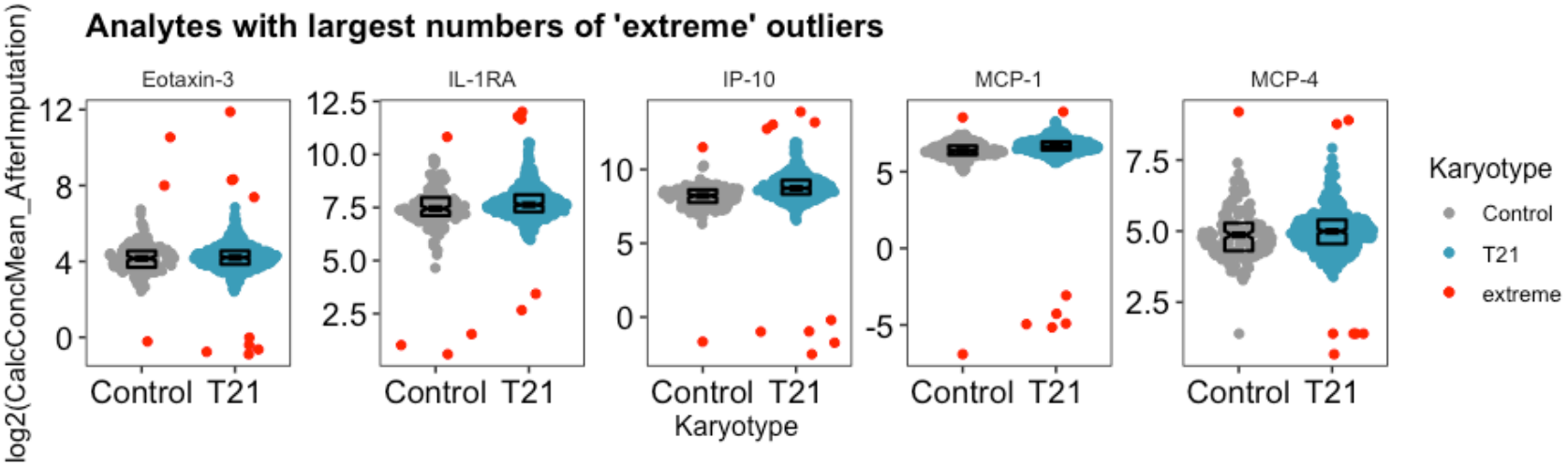
Linear regression modeling for a *whole dataset* in R

1. Check for outliers

- Extreme outlier = 3x interquartile range below or above the first and third quartiles, respectively (below Q1 – 3*IQR or above Q3 + 3*IQR)
- rstatix::identify_outliers()
  or
  mutate(extreme = rstatix::is_extreme)
- Be careful: check how many data points would be removed (per group)!

```
msd_data %>%
  inner_join(meta_data) %>%
  group_by(Analyte, Karyotype) %>%
  mutate(extreme = rstatix::is_extreme(log2(CalcConcMean_AfterImputation)))
%>%
  ungroup() %>%
  filter(extreme == TRUE) %>%
  count(Analyte, name = "n_extreme") %>%
  arrange(-n_extreme)
```

```
# A tibble: 36 × 2
   Analyte   n_extreme
   <chr>         <int>
 1 Eotaxin-3        12
 2 IP-10            11
 3 IL-1RA            9
 4 MCP-1             8
 5 MCP-4             8
 6 TARC              8
 7 VEGF-C            8
 8 MDC               7
 9 Eotaxin           6
10 IFN-gamma         6
# … with 26 more rows
```



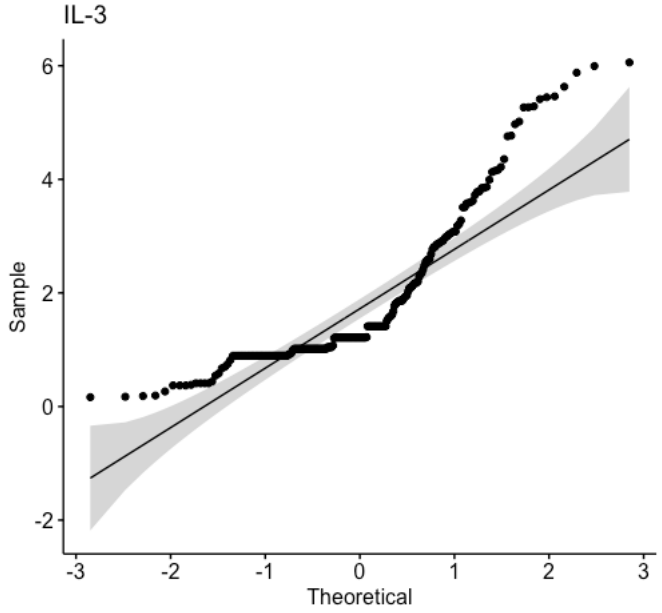
Linear regression modeling for a *whole dataset* in R

2. Check for normality

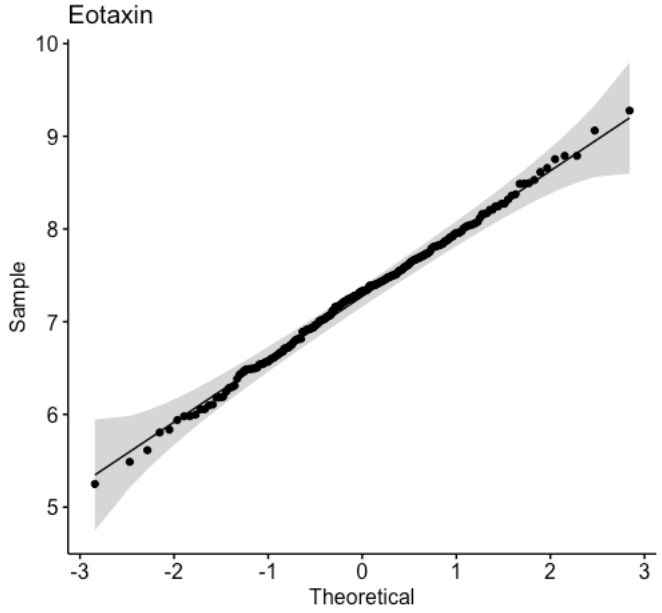
- Plot the data!!
- Shaprio-Wilk test (stringent but can identify worst offenders)
- `rstatix::shapiro_test()`
- Q-Q plots

```
msd_data_full %>%
  group_by(Analyte) %>%
  mutate(extreme = rstatix::is_extreme(log2_concentration)) %>%
  filter(extreme != TRUE) %>% # exclude extreme outliers
  group_by(Analyte) %>%
  rstatix::shapiro_test(log2_concentration) %>%
  arrange(p)

# A tibble: 54 × 4
   Analyte     variable           statistic        p
   <chr>       <chr>                  <dbl>    <dbl>
 1 IL-3        log2_concentration     0.827 3.40e-15
 2 IL-13       log2_concentration     0.880 5.13e-12
 3 VEGF-C      log2_concentration     0.905 1.33e-10
 4 IFN-gamma   log2_concentration     0.936 2.70e- 8
 5 IFN-beta    log2_concentration     0.940 4.17e- 8
 6 IL-9        log2_concentration     0.945 1.26e- 7
 7 PlGF        log2_concentration     0.946 2.07e- 7
 8 IFN-alpha2a log2_concentration     0.953 9.84e- 7
 9 TSLP        log2_concentration     0.953 1.04e- 6
10 TNF-beta    log2_concentration     0.954 1.15e- 6
# … with 44 more rows
```

"worst"



"best"



Linear regression modeling for a *whole dataset* in R

How to avoid running MANY separate regressions sing a Nest-Map-Unnest approach

3. Assemble data for regression analysis

- Select variables
- Filter outliers
- Check/filter by minimum sample number
- Check >1 level for all categorical variables
- Nest the data by feature/analyte

```
regressions_dat <- msd_data_full %>%
  select(RecordID, LabID, Sex, Age, BMI, Anti_IFNa2_titer, Sample_source, Experiment, Analyte, log2_concentration) %>%
  # filter outliers
  group_by(Analyte) %>% # using both groupings here for categorical testing
  mutate(extreme = rstatix::is_extreme(log2_concentration)) %>%
  filter(extreme != TRUE) %>% # remove extreme outliers
  ungroup() %>%
  # check minimum N
  group_by(Analyte) %>% # CHECK CORRECT GROUPING
  add_count(Sex) %>% # count by EACH categorical variable
  filter(n >= 10) %>% # require at least NN samples in each category # CURRENTLY KEEPING 23 of 30 clusters
  mutate( # count number of levels for EACH categorical variable
    # autoAb_levels = autoAb %>% fct_drop() %>% levels() %>% length(),
    Sex_levels = Sex %>% fct_drop() %>% levels() %>% length()
  ) %>%
  filter(Sex_levels > 1) %>% # need to require >1 level for each categorical level or lm() gives error
  select(-Sex_levels) %>%
  ungroup() %>%
  nest(data = c(RecordID, LabID, Sex, Age, BMI, Anti_IFNa2_titer, Sample_source, Experiment, log2_concentration, extreme, n))
#
```

Linear regression modeling for a *whole dataset* in R

Original data tibble:

```
> msd_data_full
# A tibble: 12,312 × 33
   RecordID    LabID    In_ArayaEtAl2021 ExperimentID   Analyte units concentration N_Imputed_Repli… Experiment PlateName
   <chr>       <chr>               <dbl> <chr>         <chr>   <chr>         <dbl>            <dbl>      <dbl> <chr>
 1 INVAB226VEU HTP0591A                1 MSD_P4C_09012… FGF (b… pg/mL          5.00                0          6 Plate_29…
 2 INVAB226VEU HTP0591A                1 MSD_P4C_09012… PlGF    pg/mL          6.52                0          6 Plate_29…
 3 INVAB226VEU HTP0591A                1 MSD_P4C_09012… Tie-2   pg/mL       4790.                  0          6 Plate_29…
 4 INVAB226VEU HTP0591A                1 MSD_P4C_09012… VEGF-C  pg/mL         25.4                 0          6 Plate_29…
 5 INVAB226VEU HTP0591A                1 MSD_P4C_09012… VEGF-D  pg/mL        756.                  0          6 Plate_29…
 6 INVAB226VEU HTP0591A                1 MSD_P4C_09012… VEGFR-… pg/mL         96.3                 0          6 Plate_29…
 7 INVAB226VEU HTP0591A                1 MSD_P4C_09012… Eotaxin pg/mL        181.                  0          6 Plate_2B…
 8 INVAB226VEU HTP0591A                1 MSD_P4C_09012… Eotaxi… pg/mL         20.6                 0          6 Plate_2B…
 9 INVAB226VEU HTP0591A                1 MSD_P4C_09012… IP-10   pg/mL        489.                  0          6 Plate_2B…
10 INVAB226VEU HTP0591A                1 MSD_P4C_09012… MCP-1   pg/mL        112.                  0          6 Plate_2B…
# … with 12,302 more rows, and 23 more variables: PlateBarcode <chr>, Plate_Num <chr>, script <chr>,
#   Date_exported <dbl>, Data_contact <chr>, Comments <lgl>, FamilyID <chr>, Event_name <fct>, Sex <fct>,
#   Karyotype <fct>, Age <dbl>, Sample_source <chr>, BMI <dbl>, CollabID <chr>, Anti_IFNa2_titer <dbl>,
#   Anti_IFNw_titer <dbl>, Neutralization_IFNa2_10_ng <chr>, Neutralization_IFNw_10_ng <chr>, IFNa2_auto <lgl>,
#   IFNw_auto <lgl>, autoAb <lgl>, batch <dbl>, log2_concentration <dbl>
```

Nested data tibble:

```
> regressions_dat
# A tibble: 54 × 2
   Analyte       data
   <chr>         <list>
 1 FGF (basic)   <tibble [228 × 11]>
 2 PlGF          <tibble [225 × 11]>
 3 Tie-2         <tibble [228 × 11]>
 4 VEGF-C        <tibble [220 × 11]>
 5 VEGF-D        <tibble [227 × 11]>
 6 VEGFR-1/Flt-1 <tibble [228 × 11]>
 7 Eotaxin       <tibble [223 × 11]>
 8 Eotaxin-3     <tibble [221 × 11]>
 9 IP-10         <tibble [220 × 11]>
10 MCP-1         <tibble [223 × 11]>
# … with 44 more rows
```

Linear regression modeling for a *whole dataset* in R

Using a Nest-Map-Unnest approach

4. Run linear regression per feature – all at once

- `purrr::map()`
  (see also `furrr::future_map()` for parallelization)
- `broom::tidy()`
- `broom::glance()`
- `broom::augment()`

```
# SIMPLE LINEAR REGRESSION
regressions_simple <- regressions_dat %>%
  mutate(
    fit = map(data, ~ lm(log2_concentration ~ log2(Anti_IFNa2_titer), data = .x)),
    tidied = map(fit, broom::tidy), # see ?tidy.lm
    glanced = map(fit, broom::glance), # see ?glance.lm
    augmented = map(fit, broom::augment) # see ?augment.lm
  )

# MULTPLE LINEAR REGRESSION
regressions_multi_SexAge <- regressions_dat %>%
  mutate(
    fit = map(data, ~ lm(log2_concentration ~ log2(Anti_IFNa2_titer) + Sex + Age, data = .x)),
    tidied = map(fit, broom::tidy), # see ?tidy.lm
    glanced = map(fit, broom::glance), # see ?glance.lm
    augmented = map(fit, broom::augment) # see ?augment.lm
  )
```

Linear regression modeling for a *whole dataset* in R

Access results by unnesting!

```
> regressions_simple
# A tibble: 54 x 6
   Analyte       data                fit    tidied           glanced           augmented
   <chr>         <list>              <list> <list>           <list>            <list>
 1 FGF (basic)   <tibble [228 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [228 x 9]>
 2 PlGF          <tibble [225 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [225 x 9]>
 3 Tie-2         <tibble [228 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [228 x 9]>
 4 VEGF-C        <tibble [220 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [220 x 9]>
 5 VEGF-D        <tibble [227 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [227 x 9]>
 6 VEGFR-1/Flt-1 <tibble [228 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [228 x 9]>
 7 Eotaxin       <tibble [223 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [223 x 9]>
 8 Eotaxin-3     <tibble [221 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [221 x 9]>
 9 IP-10         <tibble [220 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [220 x 9]>
10 MCP-1         <tibble [223 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [223 x 9]>
# … with 44 more rows
```

broom::tidy(): Model stats

```
> regressions_simple %>% unnest(tidied)
# A tibble: 108 x 10
   Analyte     data                fit    term                 estimate std.error statistic   p.value glanced  augmented
   <chr>       <list>              <list> <chr>                   <dbl>     <dbl>     <dbl>     <dbl> <list>   <list>
 1 FGF (basic) <tibble [228 x 11]> <lm>   (Intercept)              1.26      0.304      4.15  4.66e- 5 <tibble> <tibble>
 2 FGF (basic) <tibble [228 x 11]> <lm>   log2(Anti_IFNa2_tite…  -0.0357    0.0801    -0.445 6.57e- 1 <tibble> <tibble>
 3 PlGF        <tibble [225 x 11]> <lm>   (Intercept)              3.01      0.128     23.4   5.17e-62 <tibble> <tibble>
 4 PlGF        <tibble [225 x 11]> <lm>   log2(Anti_IFNa2_tite…   0.0116    0.0338     0.343 7.32e- 1 <tibble> <tibble>
 5 Tie-2       <tibble [228 x 11]> <lm>   (Intercept)             12.5       0.0908   138.    5.15e-220 <tibble> <tibble>
 6 Tie-2       <tibble [228 x 11]> <lm>   log2(Anti_IFNa2_tite…  -0.0185    0.0240    -0.771 4.41e- 1 <tibble> <tibble>
 7 VEGF-C      <tibble [220 x 11]> <lm>   (Intercept)              5.15      0.291     17.7   5.23e-44 <tibble> <tibble>
 8 VEGF-C      <tibble [220 x 11]> <lm>   log2(Anti_IFNa2_tite…   0.0616    0.0772     0.799 4.25e- 1 <tibble> <tibble>
 9 VEGF-D      <tibble [227 x 11]> <lm>   (Intercept)              9.22      0.123     74.8   6.01e-161 <tibble> <tibble>
10 VEGF-D      <tibble [227 x 11]> <lm>   log2(Anti_IFNa2_tite…   0.0455    0.0325     1.40  1.64e- 1 <tibble> <tibble>
# … with 98 more rows
```

Linear regression modeling for a *whole dataset* in R

Access results by unnesting!

```
> regressions_simple
# A tibble: 54 x 6
   Analyte       data                fit    tidied           glanced           augmented
   <chr>         <list>              <list> <list>           <list>            <list>
 1 FGF (basic)   <tibble [228 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [228 x 9]>
 2 PlGF          <tibble [225 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [225 x 9]>
 3 Tie-2         <tibble [228 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [228 x 9]>
 4 VEGF-C        <tibble [220 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [220 x 9]>
 5 VEGF-D        <tibble [227 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [227 x 9]>
 6 VEGFR-1/Flt-1 <tibble [228 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [228 x 9]>
 7 Eotaxin       <tibble [223 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [223 x 9]>
 8 Eotaxin-3     <tibble [221 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [221 x 9]>
 9 IP-10         <tibble [220 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [220 x 9]>
10 MCP-1         <tibble [223 x 11]> <lm>   <tibble [2 x 5]> <tibble [1 x 11]> <tibble [223 x 9]>
# … with 44 more rows
```

broom::glance():
Model metrics

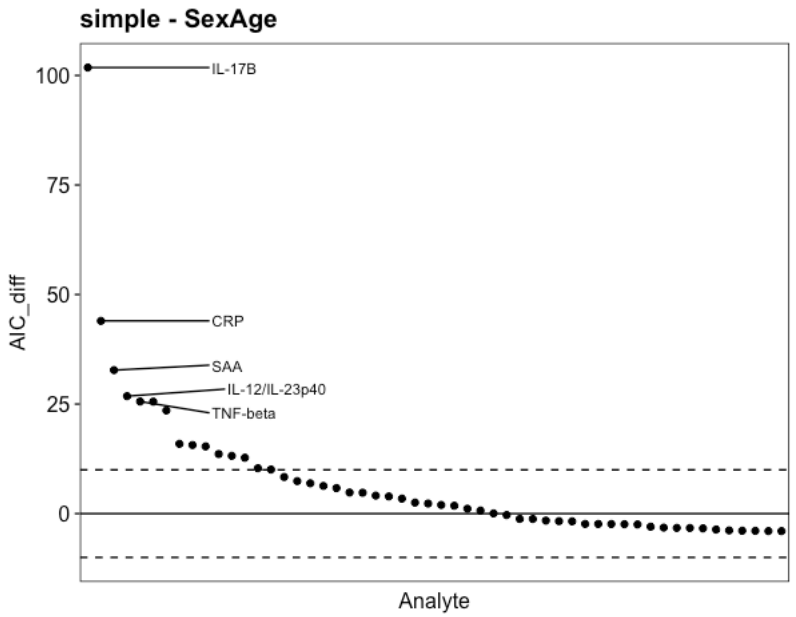
```
> regressions_simple %>% unnest(glanced)
# A tibble: 54 x 16
   Analyte      data     fit   tidied   r.squared adj.r.squared sigma statistic p.value    df logLik   AIC   BIC deviance
   <chr>        <list>   <lis> <list>       <dbl>         <dbl> <dbl>     <dbl>   <dbl> <int>  <dbl> <dbl> <dbl>    <dbl>
 1 FGF (basic)  <tibble> <lm>  <tibble>  0.000876     -0.00354  1.39     0.198    0.657     2  -397.  800.  811.    435.
 2 PlGF         <tibble> <lm>  <tibble>  0.000529     -0.00395  0.583    0.118    0.732     2  -197.  400.  410.     75.8
 3 Tie-2        <tibble> <lm>  <tibble>  0.00263      -0.00179  0.415    0.595    0.441     2  -122.  250.  260.     38.9
 4 VEGF-C       <tibble> <lm>  <tibble>  0.00292      -0.00166  1.31     0.638    0.425     2  -371.  748.  758.    375.
 5 VEGF-D       <tibble> <lm>  <tibble>  0.00861       0.00420  0.563    1.95     0.164     2  -191.  387.  398.     71.3
 6 VEGFR-1/Flt… <tibble> <lm>  <tibble>  0.000613     -0.00381  0.406    0.139    0.710     2  -117.  240.  251.     37.3
 7 Eotaxin      <tibble> <lm>  <tibble>  0.00176      -0.00276  0.692    0.389    0.534     2  -233.  473.  483.    106.
 8 Eotaxin-3    <tibble> <lm>  <tibble>  0.00849       0.00396  0.616    1.87     0.172     2  -206.  417.  427.     83.2
 9 IP-10        <tibble> <lm>  <tibble>  0.00432      -0.000244 0.846    0.946    0.332     2  -274.  555.  565.    156.
10 MCP-1        <tibble> <lm>  <tibble>  0.000113     -0.00441  0.465    0.0250   0.874     2  -145.  295.  305.     47.7
# … with 44 more rows, and 2 more variables: df.residual <int>, augmented <list>
```

Linear regression modeling for a *whole dataset* in R

5. Aikake Information Criterion (AIC) comparison

- Decrease in AIC = better model
- Threshold at 2/5/10

```
simple_glance %>% select(Analyte, AIC1 = AIC) %>%
  inner_join(multi_SexAge_glance %>% select(Analyte, AIC2 = AIC)) %>%
  mutate(AIC_diff = AIC1 - AIC2) %>%
  arrange(-AIC_diff) %>%
  mutate(Analyte = fct_inorder(Analyte)) %>%
  ggplot(aes(Analyte, AIC_diff)) +
  geom_hline(yintercept = 0) +
  geom_hline(yintercept = 10, linetype = 2) +
  geom_hline(yintercept = -10, linetype = 2) +
  geom_point() +
  theme(axis.text.x = element_blank(), axis.ticks.x = element_blank()) + # turn off
with too many
  labs(title = "simple - SexAge") +
  geom_text_repel(data = . %>%  slice_max(AIC_diff, n = 5), aes(label = Analyte), xlim
= c(10, NA), size = 3)
```



Linear regression modeling for a *whole dataset* in R

6. Assemble regression results table

- Exclude intercept rows (usually)
- Multiple hypothesis correction (line in bold below)

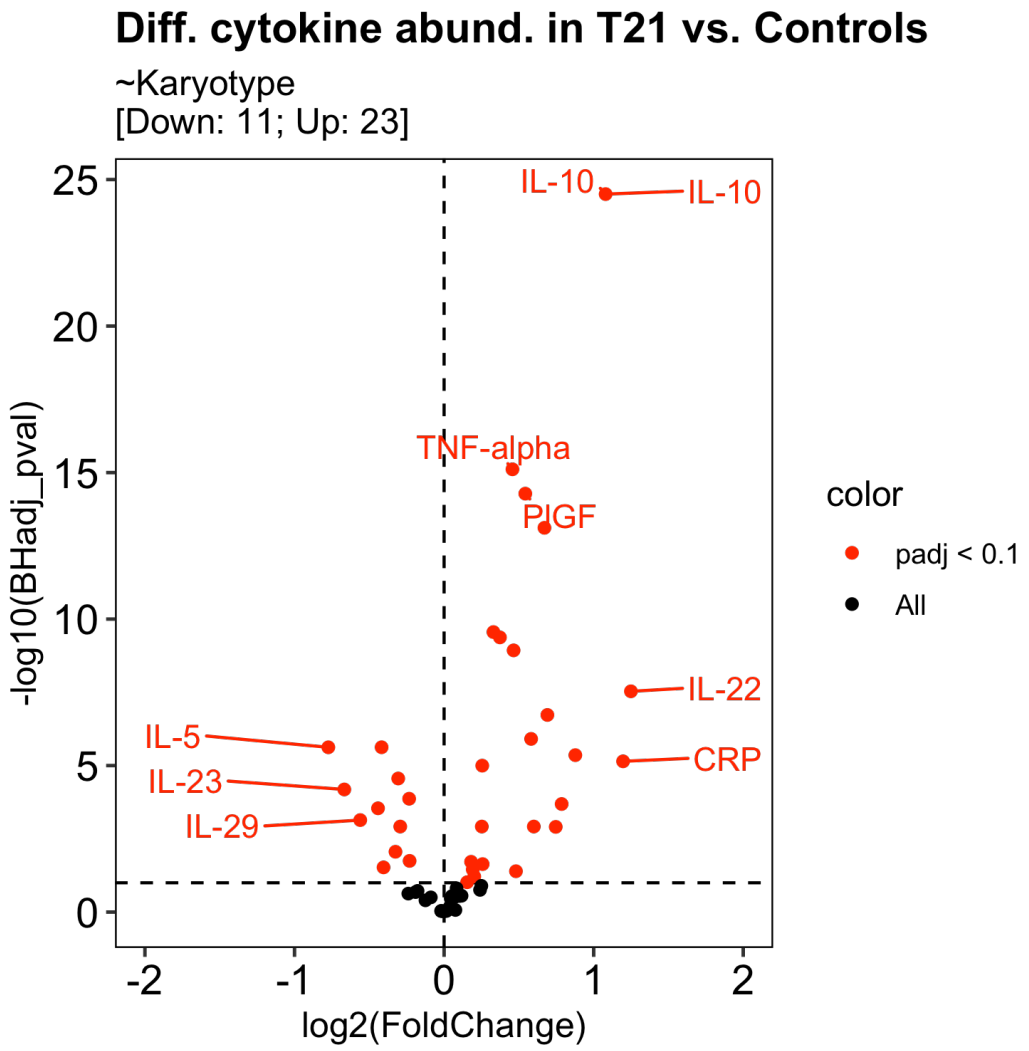
```
lm_results_simple <- regressions_simple %>%
  unnest(tidied) %>%
  select(Analyte, term, estimate, p.value) %>%
  group_by(Analyte) %>%
  dplyr::summarize(
    Analyte = first(Analyte),
    # log2_denom = first(estimate), # check for transformation and adjust accordingly
    # log2_num = nth(estimate, n = 2) + log2_denom, # check for transformation and
adjust accordingly
    log2FoldChange = nth(estimate, n = 2), # check for transformation and adjust
accordingly; equivalent to difference between level 2 and level 1 ie y = B0 + B1x
    FoldChange = 2^log2FoldChange, # check for transformation and adjust accordingly
    pval = nth(p.value, n = 2)
  ) %>%
  ungroup() %>%
  arrange(pval) %>%
  mutate(BHadj_pval = p.adjust(pval, method = "BH", n = length(pval))) %>%
  select(
    Analyte,
    # log2_denom,
    # log2_num,
    FoldChange,
    log2FoldChange,
    pval,
    BHadj_pval,
    everything()
  )
```

```
# A tibble: 108 x 4
   Analyte     term                  estimate   p.value
   <chr>       <chr>                    <dbl>     <dbl>
 1 FGF (basic) (Intercept)             1.26   4.66e-  5
 2 FGF (basic) log2(Anti_IFNa2_titer) -0.0357 6.57e-  1
 3 PlGF        (Intercept)             3.01   5.17e- 62
 4 PlGF        log2(Anti_IFNa2_titer)  0.0116 7.32e-  1
 5 Tie-2       (Intercept)            12.5    5.15e-220
 6 Tie-2       log2(Anti_IFNa2_titer) -0.0185 4.41e-  1
 7 VEGF-C      (Intercept)             5.15   5.23e- 44
 8 VEGF-C      log2(Anti_IFNa2_titer)  0.0616 4.25e-  1
 9 VEGF-D      (Intercept)             9.22   6.01e-161
10 VEGF-D      log2(Anti_IFNa2_titer)  0.0455 1.64e-  1
# … with 98 more rows
```

```
> lm_results_simple
# A tibble: 54 x 5
   Analyte     FoldChange log2FoldChange   pval BHadj_pval
   <chr>            <dbl>          <dbl>  <dbl>      <dbl>
 1 IL-2             1.12           0.169  0.0329      0.629
 2 IL-17A           1.07           0.0991 0.0369      0.629
 3 VEGF-A           0.947         -0.0782 0.0463      0.629
 4 IL-1alpha        1.13           0.174  0.0466      0.629
 5 IFN-alpha2a      1.07           0.100  0.0782      0.679
 6 IFN-beta         1.13           0.172  0.0856      0.679
 7 IL-10            1.06           0.0815 0.121       0.679
 8 IL-4             1.06           0.0831 0.130       0.679
 9 TARC             0.947         -0.0778 0.155       0.679
10 IL-16            1.03           0.0448 0.159       0.679
# … with 44 more rows
```

Linear regression modeling for a *whole dataset* in R

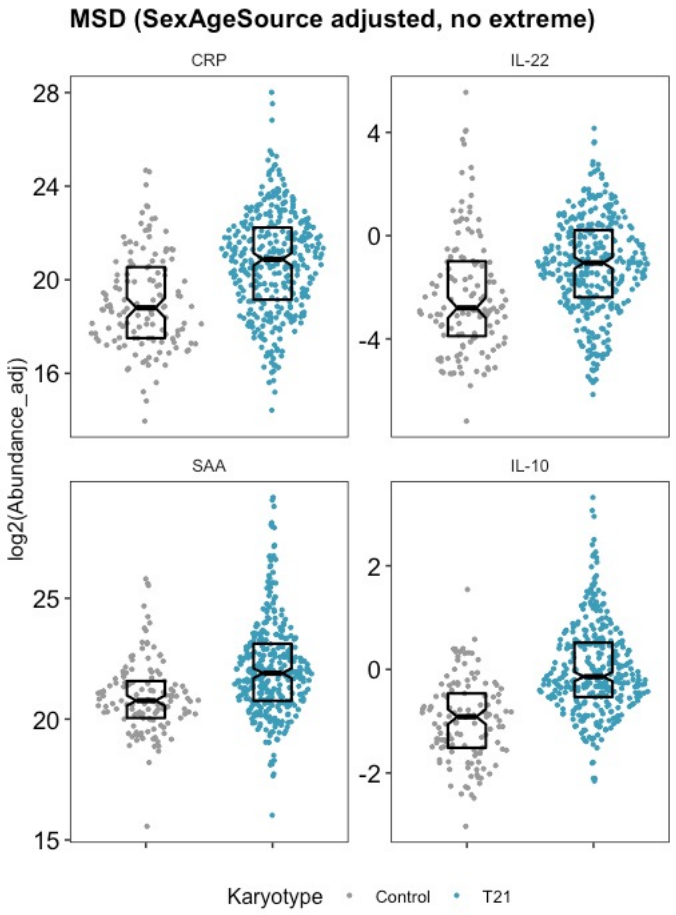
Volcano plot:



Linear regression modeling for a *whole dataset* in R

Plot individual features:

```
msd_SourceSexAge_adj %>%
  filter(Analyte %in% c("CRP", "IL-22", "SAA", "IL-10", "IL-10")) %>%
  inner_join(meta_data) %>%
  mutate( # to control order of facets
    Analyte = fct_relevel(Analyte, c("CRP", "IL-22", "SAA", "IL-10"))
    ) %>%
  group_by(Analyte, Karyotype) %>%
  mutate(extreme = rstatix::is_extreme(log2(Abundance_adj))) %>%
  ungroup() %>%
  filter(extreme == FALSE) %>%
  ggplot(aes(Karyotype, log2(Abundance_adj), color = Karyotype)) +
  geom_sina(size = 0.75) +
  geom_boxplot(notch = TRUE, varwidth = FALSE, outlier.shape = NA, coef =
FALSE, width = 0.3, color = "black", fill = "transparent", size = 0.75) +
  facet_wrap(~ Analyte, scales = "free_y", nrow = 1) +
  scale_color_manual(values = standard_colors) +
  labs(
    title = "MSD (SexAgeSource adjusted, no extreme)",
    x = NULL
  ) +
  theme(
    aspect.ratio = 1.3,
    axis.text.x = element_blank(),
    legend.position = "bottom"
  )
```



Hands on:

1. **Linear regression exercise**
2. **HTP linear regression example**

How we will use Git and GitHub

GitHub repository: https://github.com/DS3-Course/DS3_2026

Simple workflow for accessing and updating course materials for DS3 afternoon sessions

- **Clone** the course repository once at the beginning:
  `git clone https://github.com/DS3-Course/DS3_2026.git`

- **Pull** updates each day to access new content:
  `git pull`
  Alternative: direct download from github

- **Open** the appropriate `.Rproj` file for each lesson in RStudio.

- Follow instructions in README.md files and R scripts.